

View downloaded files setting download data logger.

Note: The fields are in a fixed position, regardless if the above fields are active or not. The disabled fields are empty (delimited by the separator but without data).

N°Record. View progressively the number of registered records.	A	NRECORD	n°
Date. The recording date viewing for each record.	B	DATE	ddmm/yy
Hours. Time recording viewing for each record.	C	HOURS	000000
Total positive totalizer value. Form Fields when the send flag is active on the totalizer T+.	D	U.M.	dm3
Partial positive totalizer value. Form Fields when the send flag is active on the totalizer P-.	E	T+	0
Total negative totalizer value. Form Fields when the send flag is active on the totalizer T-.	F	U.M.	dm3
Partial negative totalizer value. Form Fields when the send flag is active on the totalizer P-.	G	P-	0
Total net totalizer value. Form Fields when the send flag is active on the totalizer TN.	H	U.M.	dm3
Partial net totalizer value. Form Fields when the send flag is active on the totalizer PN.	I	T-	0
Flow rate. Form Fields present when the send flag is on the flow in units of measurement.	J	U.M.	dm3
Flow rate %. Form fields present when the flag of alarm recording is active (only N ° of present total alarms)	K	P-	0
N ° active alarms. Form fields present when the flag of alarm recording is active (only N ° of present total alarms)	L	U.M.	dm3
Loss of current measured during insulation test. Available value when recording the sensor test data is active.	M	TN	0
Time rise A. Available value when recording the sensor test data is active.	N	U.M.	dm3
Time rise B. Available value when recording the sensor test data is active.	O	PN	0
Sensor test error code. Available value when recording the sensor test data is active.	P	U.M.	dm3/s
	Q	U.M.	dm3/s
	R	U.M.	dm3/s
	S	U.M.	dm3/s
	T	U.M.	dm3/s
	U	U.M.	dm3/s
	V	U.M.	dm3/s
	W	U.M.	dm3/s
	X	U.M.	dm3/s
	Y	U.M.	dm3/s
	Z	U.M.	dm3/s
	AA	U.M.	dm3/s
	AB	U.M.	dm3/s
	AC	U.M.	dm3/s

Visualization of downloaded file. Access Level 2 (diagnostic level) is required in order to download this type of file.

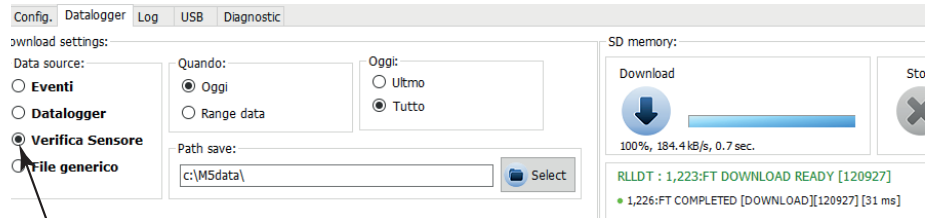
Voltage measured on electrode E1. Form fields when is active the recording of data on the input voltage (diagnostic value).	AD U.M. V 0	AE U.M. V 0
Voltage measured on electrode E2. Form fields when is active the recording of data on the input voltage (diagnostic value).	AF U.M. V -0.023	AG U.M. V -0.023
Differential voltage between the two electrodes. Form fields when is active the recording of data on the input voltage (diagnostic value).	AH U.M. V 0	AI U.M. V 0
Common mode voltage in the electrodes. Form fields when is active the recording of data on the input voltage (diagnostic value).	AJ U.M. V 0	AK U.M. V 0
Noise at low frequency measured on the electrodes. Form fields when is active the recording of data on the input signal noise levels (diagnostic value).	AL U.M. V 0	AM U.M. V 0
Differential low frequency noise measured on the electrodes. Form fields when is active the recording of data on the input signal noise levels (diagnostic values).	AN U.M. V 0	AO U.M. V 0
Low-frequency noise measured input ADC. Form fields when is active the recording of data on the input signal noise levels (diagnostic values).	AP mV 0	AQ U.M. mV 0
High frequency noise measured input ADC. Form fields when is active the recording of data on the input signal noise levels (diagnostic values).	AR mV 0	AS U.M. mV 0
Measured equivalent resistance on the electrode 1. Form fields when is active the recording of data on the electrode resistance measurements (diagnostic values).	AT U.M. kohm 0	AU U.M. kohm 0
Measured equivalent resistance on the electrode 2. Form fields when is active the recording of data on the electrode resistance measurements (diagnostic values).	AV U.M. kohm 0	AW U.M. kohm 0
Coils excitation current. Form fields when is active the recording of data related to the sensor excitation circuit measures (diagnostic value)	AX mA 0	AY U.M. mA 0
Measured resistance of the excitation circuit (coil + cable). Form fields when is active the recording of data relative to the sensor excitation circuit measures (diagnostic values).	AZ ohm 0	BA U.M. ohm 0
Temperature measured on the sensor coils (indirect measurement). Form fields when the data transmission flag is active relative to the sensor excitation circuit measures (diagnostic values).	BB °C 0	BC U.M. °C 0
Temperature T1 (sheet sensor 1). Form fields when the data transmission flag on board the internal temperature measurement is active (diagnostic values).	BD °C 0	BE U.M. °C 0
Temperature T2 (sheet sensor 2). Form fields when the data transmission flag on board the internal temperature measurement is active (diagnostic values).	BF °C 0	BG U.M. °C 0
CPU temperature. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic value).	BH °C 0	BI U.M. °C 0
Primary power supply of CPU. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic value).	BJ V 0	BK U.M. V 0
Positive supply voltage of analog circuits. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BL V 0	BM U.M. V 0
Negative supply voltage of the analog circuits. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BN V 0	BO U.M. V 0
Voltage measured on the battery B1 (NOT rechargeable battery). Fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BP V 0	BQ U.M. V 0
Voltage measured on the battery B2 (or rechargeable battery). Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BR V 0	BS U.M. V 0
% Battery charge. Form fields when the data on the card's internal power supply voltage measurements flag is ON (diagnostic values).	BT % 0	BU U.M. % 0

Example: Sensor Verify

The function "Sensor Verify" creates the "STESTLOG.CSV" file, according to the following conditions:

- 1) Activate the SDC / RTC option in group "HW Config" (by factory see the order code)
- 2) Activate the "BIV" in the group "PRODUCT CODE" (by factory see the order code)
- 3) Activate the "Sens.verify" function in the "Sensor" menu

Activated such conditions, the converter will perform every hour a measure of sensor's parameters and record a line of the file "STESTLOG.CSV"; it is possible even a manual verification by the command "sens.verify" on the menu "Diagnostic" or through the MCP command "SVERC".



Per scaricare tutte le verifiche sensore del giorno odierno in una cartella specifica, settare i seguenti parametri come segue:

Data source: Verifica sensore

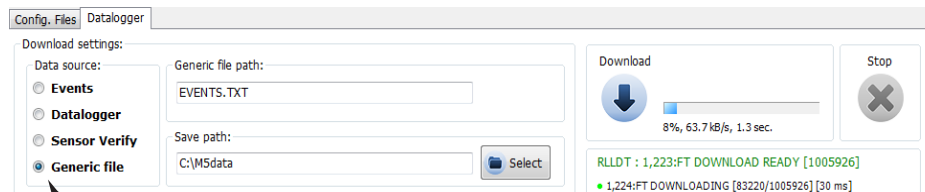
Quando: Oggi

Oggi: tutti

Save path: C:/.....

Impostati i parametri premere il tasto Download.

Example: Generic File



Questa Funzione permette di selezionare un file contenuto nella memoria SD del convertitore e scaricarlo.

Nella sezione "Generic file path" indicare il nome del file contenuto nella SD e indicare in "Save path" il percorso di salvataggio del file stesso. Impostati tali valori procedere con il download del file cliccando sul tasto "download".

MENU 11 - FUNCTION

The following functions are activated by first pressing the "ENTER" and then the "ESC" when the screen appears "confirm" to start the function.

(POS. 11.1) <i>Total direct totalizer reset</i>	[T+ RESET= ON]	AL3	[VTTPR]
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Reset total direct totalizer for direct flow rate (+)

(POS. 11.2) <i>Partial direct totalizer reset</i>	[P+ RESET= ON]	AL3	[VTPPR]
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Reset total partial totalizer for direct flow rate (+)

(POS. 11.3) <i>Total reverse totalizer reset</i>	[T- RESET= ON]	AL3	[VTTNR]
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Reset total reverse totalizer for direct flow rate (-)

(POS. 11.4) <i>Partial reverse totalizer reset</i>	[P- RESET= ON]	AL3	[VTPNR]
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Reset partial reverse totalizer for direct flow rate (-)

(POS. 11.5) <i>Load factory default sensor</i>	[Load sens.f.def= ON]	AL3	[LFDSD]
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This function resets the parameters of the sensor factory default. To Load the saved files see function (11.7).

(POS. 11.6) <i>Load factory default converter</i>	[Load conv.f.def= ON]	AL3	[LFDSD]
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This function resets the parameters of the converter factory default. o Load the saved files see function (11.8).

(POS. 11.7) <i>Save sensor factory default</i>	[Save sens.f.def= ON]	AL6	[SFDSD]
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This function save the parameters of the sensor factory default.

(POS. 11.8) <i>Save convert factory default</i>	[Save conv.f.def= ON]	AL6	[SFDSD]
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This function loads the data from a converter to another.

(POS. 11.9) <i>Calibration Immediate</i>	[Calibration]	AL5	[CALIC]
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Perform manually a board's calibration. Press Enter and the message " EXECUTE?" will be visualized on the display then press long the key Enter to proceed. Press any other key to delete the operation.

If the sensor table is valid, the calibration is performed also when one of the following parameter has been change:

1. SENSOR DIAMETER -> Menu Sensor1
2. SENSOR MODEL -> Menu Sensor1
3. Exc. CURRENT -> Menu Sensor1
4. S. Freq. -> Menu Sensor1

To check the calibration status, active or inactive, type the command MCP Calic? and check as follows:

- ☐ CALIC = 1 calibration in progress
- ☐ CALIC = 0 calibration terminated

MENU 11 - FUNCTION: ONLY MCP FUNCTIONS

<i>Save sensor reference data</i>	[MCP ONLY]	AL4	[SRFDS]
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Save conv.f.def= ON. "meter data" page 75

MENU 12 - DIAGNOSTIC

(POS. 12.1) *Self Test Diagnostic*

[Self Test]

AL3

[ATSIC]

Meter auto-test function. This function stops the normal functions of the meter and performs a complete test cycle on the measure input circuits and on the excitation generator. To activate this function, after select it, push key Enter, at the question: "CONFIRM EXEC.?" Long Push the same key to start auto-test, or any other key for delete operation. At the end of operation the converter will revert to one of the initial visualization pages. This function is automatically performed when switching on the device. This function restarts the converter.

(POS. 12.2) *Test display*

[Test display]

AL1 NO MCP COMMAND

This function allows to do a physical test the graphic display. During this operation, 4 sequences are displayed to test the correct functioning of the device.

(POS. 12.3) *Sensor verify*

[Sens. verify]

AL3

[SVERC]

This function perform a manual sensor verification (if BIV is active)

(POS. 12.4) *Flow rate simulation*

[Flow sim=ON]

AL3

[MSIEN]

Flow rate simulation enabling. With this function it is possible to generate an internal signal that simulates the flow rate, allowing the outputs and all the connected instruments test. After enabling it, a '▲▼' appears in the top left of the screen and the flow rate simulation can be:

- ☐ set: by pushing the key Enter from one of visualization pages, to set the required % flow rate (Fl. rate=%) and the same key to confirm the value;
- ☐ finished: by pushing the key Enter from visualization pages and then by long pushing the same key.

(POS. 12.5) *Diagnostic measures*

[Display measures]

AL5

DMVLS

This Function shows the values of the various internal parameters as listed below :

```

UCPU:V      5.01  LFN_COM:V      0.0000
UPS:V       5.41  LFN_DIF:V      0.0000
VUSB:V      4.53  HFN_ADC:V      0.0000
+AVCC:V     +10.1 LFN_ADC:V      0.0001
-AVCC:V      -9.9 MEAS_NB:V      0.0000
VBATT:V      0.000 CAL_I:V      16.503
IBATT:V      0.000 CAL_U:V      3230.7783
VIN1:V      -8.071 CAL_G:V      0.1633
VIN2:V      -8.065 CAL_O:V      -0.0055
VIN_C:V     -8.068 CAL_C:V      1.0000
VIN_D:V     -0.006 CAL_R0:V      1.0000
C_C:MA      25.00  CAL_R1:V      1.0000
C_U:V        5.0   CAL_R2:V      1.0000
C_UPK:V      5.5   CAL_R3:V      1.0000
C_R:Q       201.9  SVS_F:MHz      50.0051
C_PWR:W      0.126 CURR_K:V      0.0000
C_T:°C       0.00  PROC1:V      46.4
C_R1:MS      0.00  PROC2:V      7.3
C_LK:MA      0.000 PROC3:V      3.8
C-ST:        1     PROC4:V      29.8
S_UER_RSLT:  0000  PROC5:V      12.6
E1R:kΩ       0.0   CPU_T:°C      +33.12
E2R:kΩ       0.0
  
```

(POS. 12.6) *Disp. comm. Diagnostic values*

[Disp. comm. vars]

AL5

DCVLS

Create a list of diagnostic values on the instrument communication.

```

PPP link status → PPP_STATUS: NETW E_IP_HDR5: 3
MCPI socket status → MCPI_S: ESTABLISH E_IP_HDR6: 0
RxCNT: 1477149 E_IP_HDR7: 0
TxCNT: 6515456 E_IP_HDR8: 0
E_SR_LINK: 0 E_IP_HDR9: 0
E_PAKTLEN: 0 E_TCPHDR1: 0
E_NETLAYR: 0 E_TCPHDR2: 0
E_TSPAYR: 0 E_UDPHDR1: 0
E_ARPHDR1: 0 E_UDPHDR2: 0
E_ARPHDR2: 0 E_UDPHDR3: 92
E_IP_HDR1: 0 E_ICMPHDR: 0
E_IP_HDR2: 0
E_IP_HDR3: 0
E_IP_HDR4: 0
  
```

Following are the states for the PPP link and MCPI to connect the device.

PPP link status:

- "UNDT" = undetermined
- "DEAD" = dead, link down, persistent condition
- "LCP" = LCP phase, transition condition
- "AUTH" = Authentication phase, transition condition
- "IPCP" = IP and DNS addressess assign phase, transition condition
- "NETW" = network established (normal persistent condition when the link is UP)
- "TERM" = link termination request, transition condition

MCPI link status:

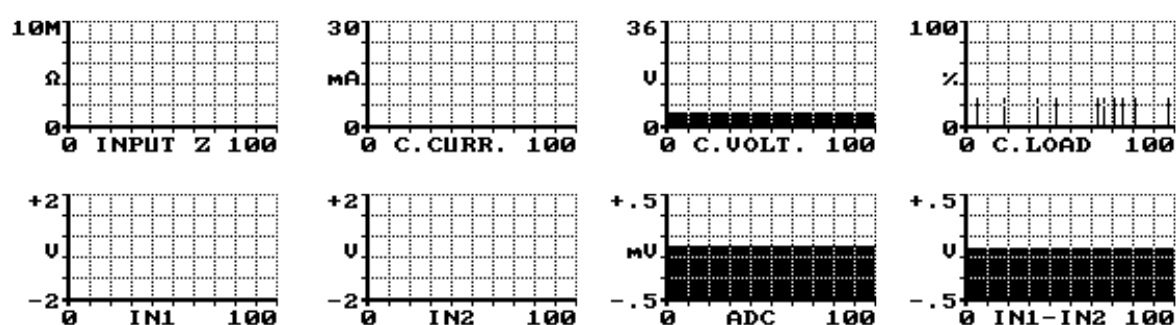
- "CLOSED" = socket closed
- "ACCEPT" = socket awaiting for new connection
- "ESTABLISH" = link established
- "CLS_WAIT" = waiting for closure
- "LAST_ACK" = lask ACK sent
- "FIN_WAIT" = (see TCP/IP RFC documentation)
- "TIME_WAIT" = (see TCP/IP RFC documentation)

(POS. 12.7) Display graphs

[Display graphs]

AL5 NO MCP COMMAND

This function displays graphs of input Z, C. current, C. Volt, C.Load, Input 1, Input 2, Input1-Input 2, Analog to Digital Converter.



(POS. 12.8) Generic sensor set

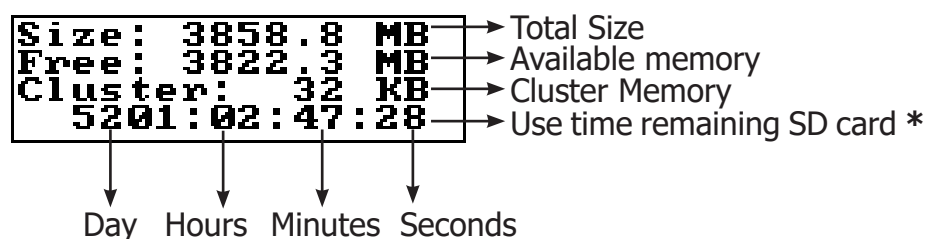
[Gen.sens. set]

AL5 NO MCP COMMAND

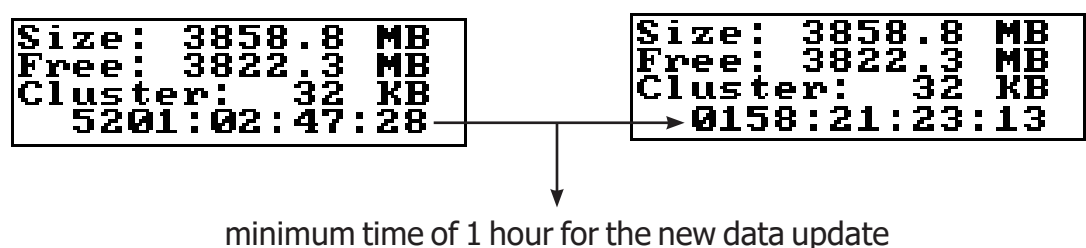
Automatic finding of a parameter set for a generic sensor.

(POS. 12.9) SD card status info. [SD card info] *ALO* **SDSTA**

This function shows the status of the SD card as shown in the following figure.



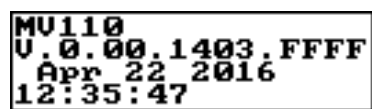
The statistical calculation is performed every time the command SDSTA is invoked. The statistical data are updated AUTOMATICALLY every day (24 hours) or manually each time the function is called to display or send the SDSTA command. However, given that for the detection is necessary a minimum of one hour, the data will not be recomputed before this time interval.



The SD card must be replaced only by the service. The use of commercial cards could disable some functionality of the instrument.

(POS. 12.10) Firmware info [Firmware info] *ALO* **MODSV**

Firmware info version/revision



(POS. 12.11) Board Serial Number [S/N= xxxxxx] *ALO* [SRNUM?]

View Board serial number. (read only)

(POS. 12.12) Working Time [WT= xxxx: xx: xx: xx] *ALO* [TWKTM?]

View Total working time instrument. (read only)

MENU 13 - SYSTEM

(POS. 13.1) Daylight saving time [Dayl. Saving= ON] *AL2* [DYSTE]

Daylight saving time change.

(POS. 13.2) Time zone [Time zone=h+xx.xx] *AL2* [TZONE]

Set time for geographic area

(POS. 13.3) Date and Time [xxxx/xx/xx-xx:xx] *AL2* [DTIME]

Set to system date and time

(POS. 13.4-5-6-7-8-9) Access level n° code [Ln xxxxxxxx] --- [L1ACD]-> [L6ACD]

This function enables or disables, for each access level code, the main menu functions.

Each level unlocks the functionality of the lower level. (Function 13.10 see page 31)

L1 code= ***** Access level value code 1 L4 code= ***** Access level value code 4

L2 code= ***** Access level value code 2 L5 code= ***** Access level value code 5

L3 code= ***** Access level value code 3 L6 code= ***** Access level value code 6

(POS. 13.10) Restricted access level [Restr. access= ON] *AL6* [RSARE]

Enable Or disable access level code. If active displays only the functions related to the level entered access.

IP ADDRESS SETTING (13.11-12-13)

(POS. 13.11) Device IP address [XXX.XXX.XXX.XXX] *AL3* [DIPAD]

Device IP network edress

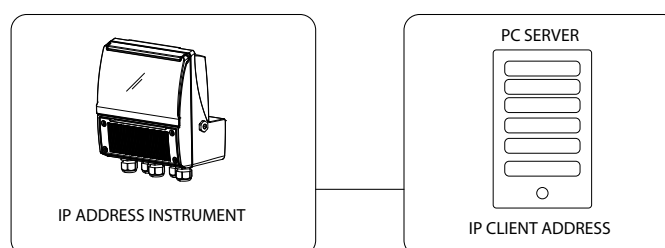
(POS. 13.12) Client IP address [XXX.XXX.XXX.XXX] *AL3* [CIPAD]

Client IP network adress

(POS. 13.13) Network mask [XXX.XXX.XXX.XXX] *AL3* [NETMS]

Network mask.

Caution: Changes to the functions of the points 13.11-13.12-13.13 are enabled after the drive device restart (see function 12.1 Self test for restart converter).



(POS. 13.14) Coefficient KT [KF=X.XXXXXX] *AL6* [CFFKT]

Gain correction coefficient (calculated automatically)

(POS. 13.15) Coefficient KS [KF=X.XXXXXX] *AL5* [CFFKS]

Correction coefficient constant instrumental

(POS. 13.16) Coefficient KR

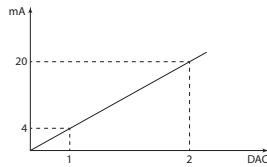
[KR=X.XXXXX]

AL5

[CFFKR]

Correction coefficient constant instrumental

DIGITAL ANALOG CONVERTER (Correction Parameters)(13.17-18-19-20)



The diagram shows how the DAC4-20mA parameters are setup. The DAC1 value corresponds to 4 mA corresponding to a zero flow rate, while the value of 20mA corresponds to a 100% of the flow rate.

(POS. 13.17) DAC1 4mA

[DAC1 4mA =XXXXX]

AL5

[C1CP1]

DAC1 out 4mA calibration point. (current output1 calibration point 1)

(POS. 13.18) DAC1 20mA

[DAC1 20mA=XXXXX]

AL5

[C1CP2]

DAC1 out 20mA calibration point. (current output1 calibration point 2)

(POS. 13.19) DAC2 4mA

[DAC1 20mA=XXXXX]

AL5

[C2CP1]

DAC2 out 4mA calibration point. (current output2 calibration point 1)

(POS. 13.20) DAC2 20mA

[DAC2 20mA=XXXXX]

AL5

[C2CP2]

DAC2 out 20mA calibration point (current output2 calibration point 2)

(POS. 13.21) Stand-BY

[STAND-BY]

AL3

[SSTBY]

Enable the converter standby state. It is enable by selecting chargeable battery in hw confing.

(POS. 13.22) Firmware Update

[FW update]

AL4

[FWUPD]

Enable firmware update. The firmware can be upload to the SD card (name.file). MCP interface is activated by the command FWUPD = name.file

MENU 13 - SYSTEM: ONLY MCP FUNCTIONS

Unique Identity KEY	[MCP ONLY]	AL0	[UIKEY]
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Device Unique Identity key

HardWare SET	[MCP ONLY]	AL0	[HWSET]
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Device hardware configuration

HardWare CODE	[MCP ONLY]	AL0	[HWCOD]
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Device hardware code

Calibr. Exec. status Memory	[MCP ONLY]	AL6	[CALXM]
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Calibration Execution status Memory. This function checks the instrument's internal calibration status.
CALXM=1 instrument calibrated correctly
CALXM=0 Invalid calibration / Calibration not completed (invalid calibration (if the function is zero, start the calibration function, MCP CALIC command)).

RTC Real time clock	[MCP ONLY]	AL2	[RTCAC]
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RTC is used to set a correction factor for the internal clock.

For setting the correction date and time with MCP press 

Function CODE Selection	[MCP ONLY]	AL0	[FCODS]
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Select the function code

FuNction Enable State Selection	[MCP ONLY]	AL6	[FNESS]
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Select the enable state of function

All FuNctions State Selection	[MCP ONLY]	AL6	[AFNSS]
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Select enable state of ALL function

Quick Start FuNction Selection	[MCP ONLY]	AL6	[QSFNS]
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Select function for quick start menu

Quick Start All Functions Selection	[MCP ONLY]	AL6	[QSAFS]
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Select ALL function converter for quick start menu.

Quick start function Status LiST	[MCP ONLY]	AL6	[QSLST]
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List quick start group functions

Function enable Status LiST	[MCP ONLY]	AL6	[FSLST]
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List enable status of functions